VPC Connection Through Transit Gateway.

1. What is AWS Transit Gateway?

AWS Transit Gateway is a **network transit hub** that connects multiple **Amazon Virtual Private Cloud (VPCs)** and **on-premise networks** using a **centralized router model**.

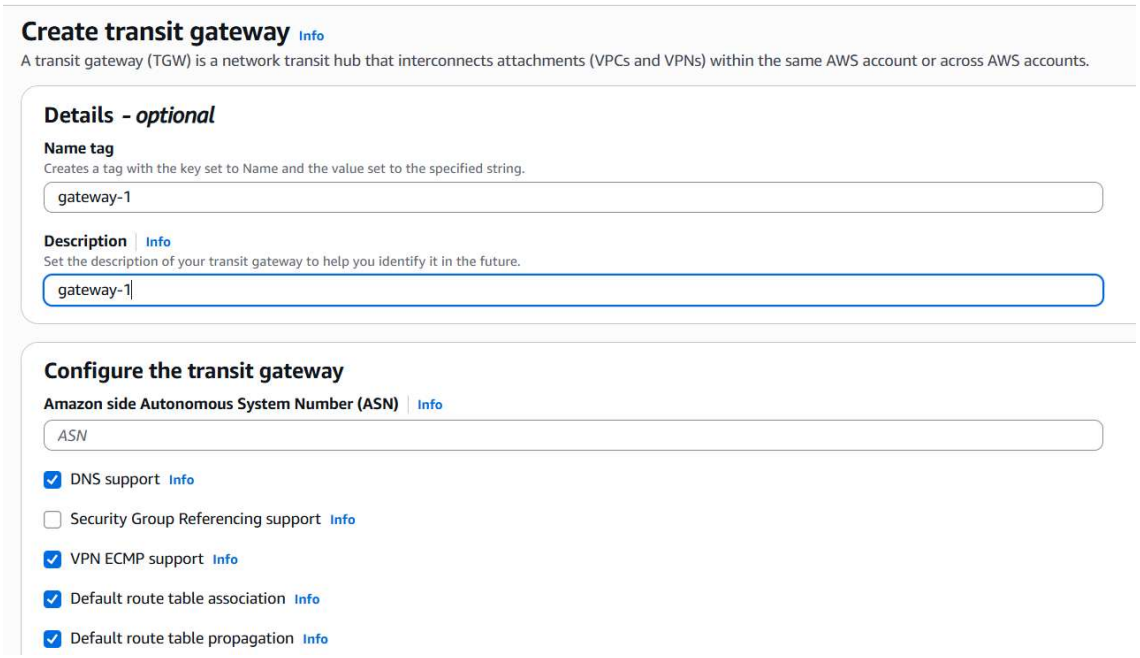
Instead of connecting VPCs individually using many **VPC Peering** connections, Transit Gateway acts like a **hub** where all networks connect to one central point.

Think of it like:

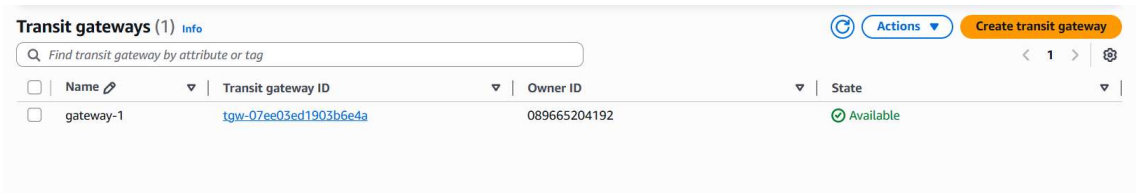
- **Transit Gateway = Router**
- **VPCs / VPN / Direct Connect = Networks connected to that router**

I have to create Transit Gateway for my three VPC's.

This is the transit gateway creation



The gateway was created successfully



This the three VPC's

Your VPCs (4) Info

Last updated 8 minutes ago | Actions | Create VPC

Find VPCs by attribute or tag

Name	VPC ID	State	Encryption c...	Encryption control ...	Block Public...	IPv4 C
–	vpc-0a3751e43faed7be1	Available	–	–	Off	172.3
vpc-2	vpc-0721c4160907c095c	Available	–	–	Off	12.0.0
vpc-3	vpc-01c1394144326851f	Available	–	–	Off	13.0.0
vpc-1	vpc-038015eda77c2e886	Available	–	–	Off	11.0.0

And then I have to create an three Transit Gateway Attachments.

attachment – 1

attachment – 2

attachment – 3

Transit gateway attachments (3) Info

Actions | Create transit gateway attachment

Find transit gateway attachment by attribute or tag

Name	Transit gateway attachment ID	Transit gateway ID	State	Resource type	Resource ID
attachment-3	tgw-attach-01d879a49851eff79	tgw-07ee03ed1903b6e4a	Available	VPC	vpc-01c1394144326851f
attachment-2	tgw-attach-069390d9696a30403	tgw-07ee03ed1903b6e4a	Available	VPC	vpc-0721c4160907c095c
attachment-1	tgw-attach-0d7fd316f97049c23	tgw-07ee03ed1903b6e4a	Available	VPC	vpc-038015eda77c2e886

I have created the three TGA for three different VPC's.

Transit gateway attachments (3) Info

Actions | Create transit gateway attachment

Find transit gateway attachment by attribute or tag

Name	Transit gateway attachment ID	Transit gateway ID	State	Resource type	Resource ID
attachment-3	tgw-attach-01d879a49851eff79	tgw-07ee03ed1903b6e4a	Available	VPC	vpc-01c1394144326851f
attachment-2	tgw-attach-069390d9696a30403	tgw-07ee03ed1903b6e4a	Available	VPC	vpc-0721c4160907c095c
attachment-1	tgw-attach-0d7fd316f97049c23	tgw-07ee03ed1903b6e4a	Available	VPC	vpc-038015eda77c2e886

The Transit Gateway one for all VPC's.

Transit gateways (1) Info

Actions | Create transit gateway

Find transit gateway by attribute or tag

Name	Transit gateway ID	Owner ID	State
gateway-1	tgw-07ee03ed1903b6e4a	089665204192	Available

Then I have edited routes for all the VPC's.

Route-1

rtb-0935ba3c3211d2947 / route-table-1

Details | Routes | Subnet associations | Edge associations | Route propagation | Tags

Routes (4)

Both | Edit routes

Filter routes

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-035ad62a258df98ee	Active	No	Create Route
11.0.0.0/16	local	Active	No	Create Route Table
12.0.0.0/16	tgw-07ee03ed1903b6e4a	Active	No	Create Route
13.0.0.0/16	tgw-07ee03ed1903b6e4a	Active	No	Create Route

Route – 2

rtb-0fd93c6e78a864c87 / route-table-2

Routes (4)

Both | Edit routes

Filter routes

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-048a8a321f1870a43	Active	No	Create Route
11.0.0.0/16	tgw-07ee03ed1903b6e4a	Active	No	Create Route
12.0.0.0/16	local	Active	No	Create Route Table
13.0.0.0/16	tgw-07ee03ed1903b6e4a	Active	No	Create Route

Route – 3

rtb-06c7c8b3dc1e12ef0 / route-table-3

Details | Routes | Subnet associations | Edge associations | Route propagation | Tags

Routes (4)

Both | Edit routes

Filter routes

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-08a8baacc5ec9f573	Active	No	Create Route
11.0.0.0/16	tgw-07ee03ed1903b6e4a	Active	No	Create Route
12.0.0.0/16	tgw-07ee03ed1903b6e4a	Active	No	Create Route
13.0.0.0/16	local	Active	No	Create Route Table

The created Route tables

Route tables (1/7) Info

Last updated less than a minute ago | Actions | Create route table

Find route tables by attribute or tag

Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
route-table-1	rtb-0935ba3c3211d2947	subnet-0e61e6329d2f33...	–	No	vpc-038015eda77c2e886 | vp
–	rtb-0a6fe5577354447f4	–	–	Yes	vpc-038015eda77c2e886 | vp
–	rtb-0cf07e6c2d169cfce	–	–	Yes	vpc-0721c4160907c095c | vp
–	rtb-053faff1e05656b85	–	–	Yes	vpc-0a3751e43faed7be1
route-table-3	rtb-06c7c8b3dc1e12ef0	subnet-0815f9a0a010c4...	–	No	vpc-01c1394144326851f | vp
route-table-2	rtb-0fd93c6e78a864c87	subnet-03beafda63a7f4e...	–	No	vpc-0721c4160907c095c | vp

Then I have checked in all the instances having internet using the ping command.

EC2 – 1

```
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-11-0-0-196 ~]$ sudo su
[root@ip-11-0-0-196 ec2-user]# ping 13.0.0.8
PING 13.0.0.8 (13.0.0.8) 56(84) bytes of data.
64 bytes from 13.0.0.8: icmp_seq=1 ttl=126 time=1.43 ms
64 bytes from 13.0.0.8: icmp_seq=2 ttl=126 time=1.02 ms
64 bytes from 13.0.0.8: icmp_seq=3 ttl=126 time=0.924 ms
64 bytes from 13.0.0.8: icmp_seq=4 ttl=126 time=0.926 ms
^C
--- 13.0.0.8 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3051ms
rtt min/avg/max/mdev = 0.924/1.073/1.429/0.208 ms
[root@ip-11-0-0-196 ec2-user]# ping 12.0.0.97
PING 12.0.0.97 (12.0.0.97) 56(84) bytes of data.
64 bytes from 12.0.0.97: icmp_seq=1 ttl=126 time=0.656 ms
64 bytes from 12.0.0.97: icmp_seq=2 ttl=126 time=0.387 ms
64 bytes from 12.0.0.97: icmp_seq=3 ttl=126 time=0.367 ms
64 bytes from 12.0.0.97: icmp_seq=4 ttl=126 time=0.347 ms
^C
--- 12.0.0.97 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3112ms
rtt min/avg/max/mdev = 0.347/0.439/0.656/0.125 ms
[root@ip-11-0-0-196 ec2-user]#
```

EC2 – 2

```
[ec2-user@ip-12-0-0-97 ~]$ sudo su
[root@ip-12-0-0-97 ec2-user]# ping 13.0.0.8
PING 13.0.0.8 (13.0.0.8) 56(84) bytes of data.
64 bytes from 13.0.0.8: icmp_seq=1 ttl=126 time=1.21 ms
64 bytes from 13.0.0.8: icmp_seq=2 ttl=126 time=0.920 ms
64 bytes from 13.0.0.8: icmp_seq=3 ttl=126 time=1.16 ms
64 bytes from 13.0.0.8: icmp_seq=4 ttl=126 time=0.964 ms
^C
--- 13.0.0.8 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3063ms
rtt min/avg/max/mdev = 0.920/1.062/1.206/0.122 ms
[root@ip-12-0-0-97 ec2-user]# ping 11.0.0.196
PING 11.0.0.196 (11.0.0.196) 56(84) bytes of data.
64 bytes from 11.0.0.196: icmp_seq=1 ttl=126 time=1.18 ms
64 bytes from 11.0.0.196: icmp_seq=2 ttl=126 time=0.331 ms
64 bytes from 11.0.0.196: icmp_seq=3 ttl=126 time=0.372 ms
64 bytes from 11.0.0.196: icmp_seq=4 ttl=126 time=0.336 ms
64 bytes from 11.0.0.196: icmp_seq=5 ttl=126 time=0.359 ms
```

i-0335ea30ec5d119d1 (server-2)

PublicIPs: 13.204.156.241 PrivateIPs: 12.0.0.97

EC2 – 3

```
        _/m/'
[ec2-user@ip-13-0-0-8 ~]$ sudo su
[root@ip-13-0-0-8 ec2-user]# ping 12.0.0.97
PING 12.0.0.97 (12.0.0.97) 56(84) bytes of data.
64 bytes from 12.0.0.97: icmp_seq=1 ttl=126 time=2.19 ms
64 bytes from 12.0.0.97: icmp_seq=2 ttl=126 time=0.911 ms
64 bytes from 12.0.0.97: icmp_seq=3 ttl=126 time=0.978 ms
64 bytes from 12.0.0.97: icmp_seq=4 ttl=126 time=0.930 ms
64 bytes from 12.0.0.97: icmp_seq=5 ttl=126 time=4.67 ms
64 bytes from 12.0.0.97: icmp_seq=6 ttl=126 time=0.984 ms
64 bytes from 12.0.0.97: icmp_seq=7 ttl=126 time=0.950 ms
^C
--- 12.0.0.97 ping statistics ---
7 packets transmitted, 7 received, 0% packet loss, time 6008ms
rtt min/avg/max/mdev = 0.911/1.659/4.673/1.302 ms
[root@ip-13-0-0-8 ec2-user]# ping 11.0.0.196
PING 11.0.0.196 (11.0.0.196) 56(84) bytes of data.
64 bytes from 11.0.0.196: icmp_seq=1 ttl=126 time=1.85 ms
64 bytes from 11.0.0.196: icmp_seq=2 ttl=126 time=0.931 ms
```

i-0a3d1323e5f765a5f (server-3)

PublicIPs: 3.7.55.164 PrivateIPs: 13.0.0.8